

CSA 1718 – Artificial Intelligence

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Prolog Program for Forward Chaining

Aim

To write and implement a Prolog program to demonstrate the concept of forward chaining.

Objective

- To understand forward chaining.
- To derive new facts from existing facts and rules.
- To implement a simple rule-based inference system.

Theory

Forward chaining starts with known facts and applies rules to derive new facts.

Example:

Fact: It is raining.

Rule: If it rains, then the ground is wet.

Conclusion: The ground is wet.

The reasoning proceeds from the available facts toward the conclusion.

Algorithm

1. Define the initial fact.
2. Define rules based on the fact.
3. Apply the rules.
4. Derive the new fact.
5. Query the derived fact.

Program

fact(rain).

rule(wet_ground) :-

```
fact(rain).
```

```
rule(umbrella) :-
```

```
    fact(rain).
```

```
forward(X) :-
```

```
    rule(X).
```

Query

```
forward(wet_ground).
```

Output

```
true.
```

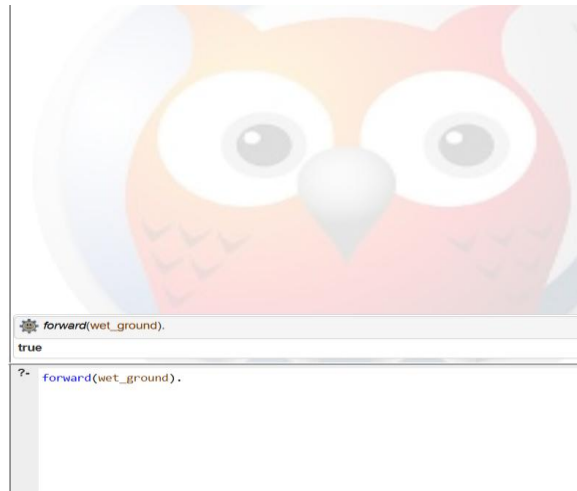
Another query:

```
forward(umbrella).
```

Output:

```
true.
```

```
1 fact(rain).  
2  
3 rule(wet_ground) :-  
4     fact(rain).  
5  
6 rule(umbrella) :-  
7     fact(rain).  
8  
9 forward(X) :-  
10    rule(X).
```



Result

Thus, the concept of forward chaining was successfully demonstrated using Prolog.